Software Project Management Business Case

For

AuralAid: Adaptive AI Hearing with Contextual Understanding

Prepared by

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1. Business Case

This business case outlines the justification for developing AuralAid: Adaptive AI Hearing with Contextual Understanding, an AI powered assistive hearing system. It evaluates different solution options, analyzes costs and benefits, assesses risks, and justifies the selected solution based on feasibility, impact, and long term value. The project aims to deliver a socially impactful, technically advanced, and scalable healthcare-assistive technology solution.

1.1 Document Information

Table 1: Document Information

Project Name:	AuralAid: Adaptive AI Hearing with Contextual Understanding
Date:	05-Nov-2025
Author:	Hamza Moeen
Approver:	Farwah Aizaz
Document Code:	DocumentCode-001
Version:	v1.0

1.2 Approval

Table 2: Approval Information

Date:	Name:	Signature:
02-Nov-2025	Junaid Ali Khan	

1.3 Notes

AuralAid aims to provide an intelligent AI-based hearing assistance system that helps hearing-impaired users understand their environment through real-time sound recognition, speech-to-text conversion, vibration alerts, and contextual AI interpretation. The system integrates embedded hardware, AI models, mobile application development, and IoT communication, addressing a real-world healthcare accessibility problem. The project serves as a Final Year Project (FYP) and aligns with both academic objectives and real-world industry relevance.

1.4 Reasons

1.4.1 Project Purpose

The purpose of AuralAid is to create an AI-powered assistive system that enhances environmental awareness for hearing-impaired individuals by converting sound into meaningful visual and haptic feedback. The project focuses on accessibility, inclusivity, and intelligent automation using AI and embedded systems.

1.4.2 Problem Statement

Hearing-impaired individuals face significant challenges in understanding environmental sounds, speech, and emergency alerts. Traditional hearing aids amplify sound but do not provide contextual understanding. There is a lack of affordable, intelligent, AI-driven assistive systems that can interpret sound meaningfully and provide actionable feedback to users.

1.5 Options

Three solution options were considered before selecting the final project direction.

1.5.1 Basic Sound Alert System

- **Description:** A simple sound detection system that triggers vibration alerts when sound is detected.
- **Advantages:** Low development cost, simple implementation, minimal AI complexity.
- **Drawbacks:** No contextual understanding, no speech recognition, limited real-world usefulness.

1.5.2 Mobile App-Based AI Assistant

- **Description:** A mobile-only application using AI models to detect sound and convert speech to text.
- **Advantages:** No hardware dependency, easier deployment, scalable software model.
- **Drawbacks:** Limited real-time accuracy, high battery usage, dependency on mobile microphones, reduced reliability.

1.5.3 AI + Embedded Smart Hearing System (AuralAid)

- **Description:** An integrated system using MEMS microphones, ESP32-S3, AI models, DSP processing, and a Flutter mobile app to provide real-time contextual sound understanding.
- **Advantages:** Real-time processing, wearable solution, intelligent feedback, scalable architecture, healthcare impact.
- **Drawbacks:** Higher technical complexity, hardware dependency, integration challenges.

1.6 Chosen Solution

AuralAid was selected as the final solution because it provides the most complete, impactful, and scalable approach. It combines AI, embedded systems, and mobile technology to deliver a real-world healthcare assistive solution. Despite higher technical complexity, it offers long-term value, innovation, and strong academic and industrial relevance.

1.7 Costs and Benefits

1.7.1 Duration

Estimated project duration: 9–10 months, including research, design, development, testing, integration, and deployment.

1.7.2 Benefits Timescale

Initial benefits expected within 3–6 months after system completion through testing, validation, academic evaluation, and demonstration.

1.7.3 Major Risks

It involves major risks are:

- AI model optimization on low resource hardware
- Hardware-software integration complexity
- Real-time processing limitations
- Testing constraints

1.7.4 Cost

Project costs are estimated at PKR 200,000, itemized as follows:

- **Hardware Components:** PKR 70,000
  (ESP32-S3, MEMS microphones, vibration motors, sensors, power modules, wearable casing)
- **Software Licenses & Tools:** PKR 50,000
  (AI development tools, model libraries, mobile app frameworks, development environments)
- **Development & Integration:** PKR 40,000
  (Embedded system integration, AI model optimization, mobile app development)
- **Testing and Validation:** PKR 40,000
  (Quality assurance, real-user testing, accessibility testing, system validation)

1.7.5 Dis-benefits

- **High Technical Complexity:**
  Integration of AI, embedded systems, mobile applications, and real-time processing increases system complexity.
- **Longer Development Time for AI Optimization:**
  Training and optimizing AI models for low-resource hardware requires extensive experimentation and tuning.
- **Hardware Dependency:**
  System performance depends on hardware quality, availability, and reliability.

1.7.6 Expected Benefits

- **Improved Accessibility & Quality of Life:**
  AuralAid enables hearing-impaired users to understand their environment through AI-based sound interpretation.
- **Portfolio & Research Value:**
  Demonstrates skills in AI, IoT, embedded systems, mobile development, and healthcare technology.
- **Social Impact:**
  Contributes to inclusive technology and assistive healthcare solutions.
- **Scalable Platform:**
  System can be expanded into smart cities, healthcare monitoring, and assistive AI ecosystems.
- **Innovation Potential:**
  Strong foundation for future research, publications, and commercialization.

1.8 Investment Appraisal

1.8.1 Combined Costs

- **Total project cost:** PKR 200,000
- **Maintenance costs:** Estimated at PKR 15,000 annually (updates, bug fixes, AI model improvements)
- **Overall financial risk:** Low (academic project scope)
- **Technical risk:** Medium (AI + hardware integration complexity)

1.8.2 Funding Justification

AuralAid offers long-term academic, social, and technological benefits. The project contributes to healthcare accessibility, AI innovation, and real-world problem solving. It builds strong interdisciplinary skills in AI, embedded systems, IoT, mobile development, and healthcare technology, making the investment highly valuable for student learning, research contribution, and future career development.

The investment of PKR 200,000 is justified due to:
• Strong academic value
• Healthcare impact
• Research potential
• Innovation contribution
• Portfolio development
• Real-world applicability

1.9 Risk Analysis

The following are the identified risks along with their corresponding mitigation strategies.

1.9.1 Development Delays

Follow modular development with milestone planning, agile sprints, and weekly progress reviews to maintain timeline discipline.

1.9.2 Budget Constraints

Use open-source AI tools, optimized hardware selection, and cost monitoring with phased spending.

1.9.3 Technical Challenges in AI & Hardware Integration

Early prototyping, lightweight AI models, faculty mentorship, and continuous integration testing.

1.9.4 User Acceptance and Reliability

Real-user testing, accessibility feedback loops, UI/UX refinement, and reliability testing in real environments.

1.10 Cost Benefit Analysis

The Cost-Benefit Analysis (CBA) assesses the financial, technical, and social advantages of proceeding with AuralAid by comparing potential solutions and quantifying their benefits and costs. The purpose is to ensure that the investment aligns with both strategic objectives and social impact goals.

1.10.1 Basic Sound Alert System (Option 1)

1. Define Problem and Objectives

- **Problem**: Limited usefulness due to lack of AI intelligence.
- **Objective**: Provide basic sound detection with vibration alerts

2. Identify Costs and Benefits

Costs (Per Year):

- Basic hardware
- Simple microcontroller
- Minimal software

Benefits (Per Year):

- Basic safety alerts
- Low maintenance cost

3. Monetary values of Costs and Benefits

Table 3: Monetary Values for Option 1

Category	Description	Value (PKR)
Hardware Components	Basic microcontroller and sensors	70,000

Software Tools	Basic development tools	15,000
Integration	Simple system integration	10,000
Testing	Basic testing	20,000
Total Costs		**115,000**
Safety Value	Basic alert system value	90,000
Accessibility Impact	Basic assistive benefit	40,000
Service Potential	Limited service usability	30,000
Total Benefits		**160,000**

4. Calculate PV of Costs and Benefits

Table 4: PV for Option 1

Year	Cost (PKR)	Benefit (PKR)	PV of Cost (PKR)	PV of Benefit (PKR)
1	115,000	160,000	109,523.81	152,380.95
2	115,000	160,000	104,308.39	145,124.72
3	115,000	160,000	99,341.32	138,213.07
4	115,000	160,000	94,610.78	131,631.50
5	115,000	160,000	90,105.50	125,363.34

5. Calculate Total PV of Costs and Benefits

- Total PV of Costs = 497,889.80 PKR
- Total PV of Benefits = 692,713.58 PKR

6. Calculate NPV

- NPV = 194,823.78 PKR

1.10.2 Mobile AI Hearing App (Option 2):

1. Define Problem and Objectives

- **Problem**: Limited reliability without dedicated wearable hardware.
- **Objective**: Provide AI-based sound recognition via mobile platform only.

2. Identify Costs and Benefits

Costs (Per Year):

- AI model development
- Cloud services
- App development
- Testing & deployment.

Benefits (Per Year):

- AI-based accessibility
- Software scalability
- Service monetization potential

3. Monetary values of Costs and Benefits

Table 5: Monetary Values for Option 2

Category	Description	Value (PKR)
AI Development	Model training and optimization	120,000
Cloud Services	AI hosting and processing	60,000
App Development	Mobile application development	50,000
Testing & QA	System testing and validation	40,000
Total Costs		**270,000**
AI Services	AI-based assistive value	200,000
Accessibility Services	Healthcare support value	120,000
Scalability Value	Expansion potential	60,000
Total Benefits		**380,000**

4. Calculate PV of Costs and Benefits

Table 6: PV for Option 2

Year	Cost (PKR)	Benefit (PKR)	PV of Cost (PKR)	PV of Benefit (PKR)
1	270,000	380,000	257,142.86	361,904.76

2	270,000	380,000	244,897.96	344,671.20
3	270,000	380,000	233,236.15	328,258.28
4	270,000	380,000	222,129.67	312,626.93
5	270,000	380,000	211,552.07	297,740.89

5. Calculate Total PV of Costs and Benefits

- Total PV of Costs = 1,168,958.71 PKR
- Total PV of Benefits = 1,645,202.06 PKR

6. Calculate NPV

- NPV = 476,243.35 PKR

1.10.3 AuralAid – AI Smart Hearing System (Chosen Option):

1. Define Problem and Objectives

- **Problem**: Complexity of AI + hardware + real-time processing integration.
- **Objective**: Develop a complete intelligent assistive hearing ecosystem.

2. Identify Costs and Benefits

Costs (Per Year):

- Hardware system
- Embedded systems
- AI model development
- Mobile application
- Testing & validation

Benefits (Per Year):

- Healthcare accessibility
- Assistive technology value
- Research contribution

- Scalability
- Commercialization potential

3. Monetary values of Costs and Benefits

Table 7: Monetary Values for Chosen Option

Category	Description	Value (PKR)
Hardware System	Wearable device & components	70,000
AI Development	Model training and optimization	60,000
Software Tools	AI tools and app frameworks	40,000
Integration	System integration	30,000
Testing & Validation	QA and real-user testing	30,000
Total Costs		**230,000**
Healthcare Value	Assistive healthcare impact	160,000
Accessibility Value	Social inclusion impact	100,000
Research Value	Academic and research contribution	60,000
Scalability Value	Expansion potential	40,000
Total Benefits		**360,000**

4. Calculate PV of Costs and Benefits

Table 8: PV for Chosen Option

Year	Cost (PKR)	Benefit (PKR)	PV of Cost (PKR)	PV of Benefit (PKR)
1	230,000	360,000	219,047.62	342,857.14
2	230,000	360,000	208,616.78	326,530.61
3	230,000	360,000	198,682.65	311,934.87
4	230,000	360,000	189,221.57	297,080.83
5	230,000	360,000	180,211.02	283,886.51

5. Calculate Total PV of Costs and Benefits

- Total PV of Costs = 995,779.64 PKR
- Total PV of Benefits = 1,562,289.96 PKR

6. Calculate NPV

- NPV = 566,510.32 PKR

1.11 Interpretation of CBA

AuralAid is the most viable and impactful option with the highest Net Present Value (NPV) and the strongest social, healthcare, and technological return on investment.

Unlike basic or app-only solutions, AuralAid delivers:
• Real-world healthcare impact
• Assistive technology value
• Research and innovation contribution
• Scalability potential
• Commercialization readiness

This makes AuralAid the optimal choice for a project focused on healthcare innovation, social impact, long-term sustainability, and real-world applicability, not just financial profitability.